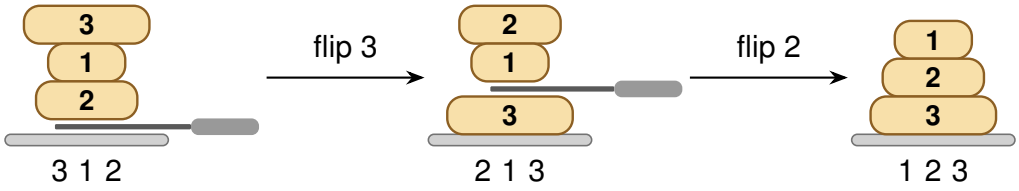
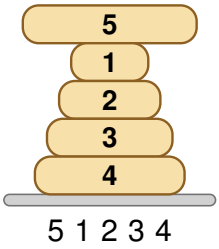
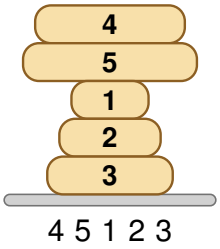
A stack is sorted when each pancake sits on a bigger one. To flip, slide the spatula under any pancake and turn that pancake and everything above it over together. Write a stack as its numbers from the top, and a flip as the number of pancakes it turns over.



Problem 1: Sort each stack with as few flips as you can. Write down your flips.



flips: _____



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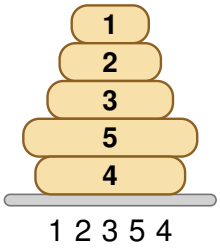

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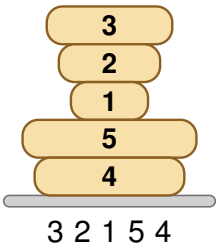

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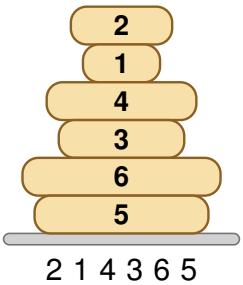
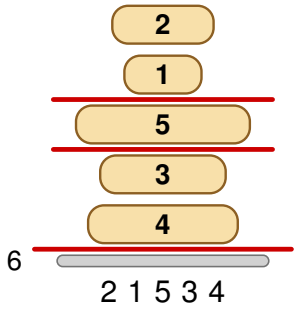


flips: _____

Problem 2: Find a way to sort any stack of six pancakes, however mixed up it is. What is the most flips your way can need? Explain why it never needs more. How many flips can your way need for a stack of 20 pancakes?

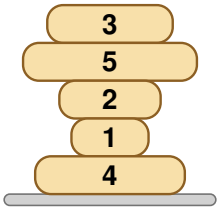
Problem 3: Think of each stack as sitting on a plate, and give the plate the number one bigger than the biggest pancake. A break is a place where two touching pancakes, or the bottom pancake and the plate, have numbers that differ by more than 1, such as 1 on 5, or 4 on plate 6 in the stack below.



2 1 5 3 4

3 breaks

Sort 3 5 2 1 4 with as few flips as you can. It has four breaks. Explain why a stack with four breaks can never be sorted in fewer than four flips.



3 5 2 1 4

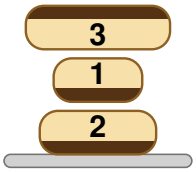
flips: _____

Problem 4: Find every stack of pancakes 1, 2, 3 and 4 that cannot be sorted in three flips or fewer. How do you know that each one cannot be sorted in three, and that you have found them all?

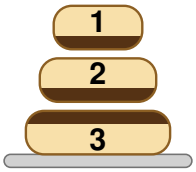
Problem 5: Find a stack of five pancakes that needs the most flips. Explain why it cannot be sorted in fewer.

Problem 6: Stack 4 6 2 5 1 3 has six breaks. Can it be sorted in six flips? If it can, show how. If it cannot, explain why.

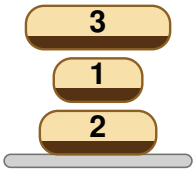
Problem 7: Now each pancake has a dark side, which must end facing down. Sort each stack with as few flips as you can.



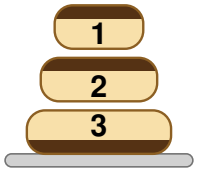
flips: _____



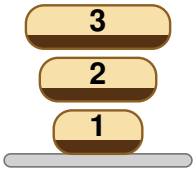
flips: _____



flips: _____

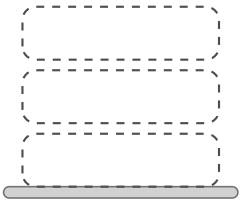


flips: _____

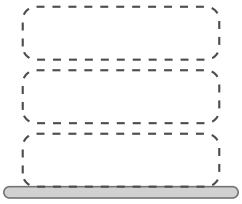


flips: _____

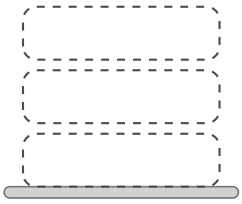
Problem 8: Find a stack of three pancakes with dark sides that needs the most flips. Draw it with its dark sides.



flips: _____



flips: _____



flips: _____

Problem 9: Find a way to sort any stack of pancakes with dark sides. What is the most flips your way can need for a stack of six?